

Xuedong Pan

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Education

Northeastern University , San Jose, CA	Sep 2023 - May 2025
Master of Science in Computer Science, GPA: 4.0/4.0	
Related Coursework: Cloud Computing, Computer Networks, Machine Learning, Computer Vision	
San Jose State University , San Jose, CA	Aug 2019 - May 2023
Bachelor of Science in Computer Science, Minor in Mathematics, GPA: 3.99/4.0, Summa Cum Laude	
Related Coursework: Data Structures and Algorithms, Operating Systems, Database Management, Object-Oriented Design	

Skills

Programming Languages: C#, Python, Java, JavaScript/TypeScript, HTML/CSS, SQL

Frameworks: .NET, Entity Framework, Flask, Django, Angular, React

Concepts & Tools: RESTful APIs, Git, Docker, SQL Server, Linux, Identity Protocols (SAML/OAuth), RBAC, SSO,

Experience

University of Iowa , Iowa City, IA	Jun 2025 - Present
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Software Engineer (Application Programmer)

- Designed and engineered the "Cambus Ridership" application using .NET 8 Web API and Entity Framework Core, supporting daily reporting workflows for campus-wide transit operations and federally mandated compliance submissions.
- Owned end-to-end system development, translating ambiguous compliance requirements into a scalable backend architecture integrated with SQL Server, ensuring long-term maintainability and auditability.
- Built security-focused audit logging for sensitive operations, reducing operational risk and enabling faster root-cause analysis during incident investigations.
- Executed a seamless data migration strategy, transforming and validating over 100,000 historical records (spanning 1985–Present) from legacy systems into a modern SQL Server architecture.
- Improved system efficiency by developing a client-side CSV export feature in TypeScript, reducing server load and eliminating external library dependencies.

Applied Medical, Rancho Santa Margarita, CA

May 2022 - Aug 2022, May 2024 - Aug 2024

Software Engineer Intern

- Architected a global exception handling middleware for the Learning Management System (LMS) using C#/.NET 6, enabling runtime error diagnosis across production and QA environments.
- Remediated critical memory leak vulnerabilities by refactoring the codebase to remove unstable dependencies (TinyMapper); implemented manual object mapping strategies to enhance application stability and performance.
- Resolved complex logic issues in the Controlled Docs System, significantly improving document search accuracy and user assignment workflows for administrators.
- Increased backend code coverage from 20% to 80% by authoring 500+ unit tests (xUnit) and optimizing CI/CD pipelines in Azure DevOps.

Projects

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| Navigation Assistance System for Visually Impaired Individuals | Oct 2024 - Dec 2024 |
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- Engineered a real-time object detection system on edge devices using Python and YOLOv11, achieving 20+ FPS inference speed for 42+ obstacle classes.
 - Integrated Stereo Vision algorithms (SGBM) to calculate precise depth/distance, processing dual-camera inputs to generate real-time audio alerts.
 - Optimized image processing pipelines to minimize latency, demonstrating strong capability in system performance optimization.

Achievements

- Awarded SJSU president's scholar in 2020, 2021, and 2022.